



## AI Playbook and Toolkit for Public Health Departments

# AI for Social Good

INT 190

AI for Social Good refers to the intentional use of artificial intelligence to improve public well-being, reduce inequities, strengthen institutions, and address complex social problems. For public health, this concept is not abstract. Public health agencies already work to prevent disease, protect communities, improve access, and reduce inequities. AI for Social Good should therefore be judged by whether it advances public health values in measurable, accountable, and community-centered ways. This training helps learners distinguish socially beneficial AI from AI that is merely innovative or efficient. A tool can be technically impressive and still fail to advance the public good if it increases surveillance burden, excludes communities, weakens transparency, or diverts resources from more effective interventions. Conversely, a modest AI tool may create meaningful public value if it reduces staff burden, improves language access, directs services to underserved communities, or helps agencies respond faster and more fairly. The module emphasizes public purpose, equity by design, community engagement, responsible innovation, transparency, measurement, and avoidance of unintended harm.

Learners will develop an AI for Social Good review brief for a public health use case.

|                   |                                             |
|-------------------|---------------------------------------------|
| Level             | introductory                                |
| Primary track     | Shared Foundational / Introductory Course   |
| Appears in tracks | shared-foundational, policy, communications |

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## Learning Objectives

- Define AI for Social Good in public health terms.
- Distinguish public benefit from novelty, efficiency, or automation alone.
- Identify how AI can support public health values, including equity, transparency, accessibility, accountability, and community trust.
- Assess potential harms, including exclusion, surveillance burden, stigma, resource diversion, and unequal benefit.
- Design community-centered safeguards for socially beneficial AI use.
- Produce an AI for Social Good review brief for a public health use case.

## Module Overview

AI for Social Good refers to the intentional use of artificial intelligence to improve public well-being, reduce inequities, strengthen institutions, and address complex social problems. For public health, this concept is not abstract. Public health agencies already work to prevent disease, protect communities, improve access, and reduce inequities. AI for Social Good should therefore be judged by whether it advances public health values in measurable, accountable, and community-centered ways.

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## Why This Topic Matters for Public Health AI

Public health is uniquely positioned to use AI for social good because its mission is population well-being rather than private gain. AI can help agencies identify emerging needs, improve access to information, reduce administrative burden, support community outreach, and allocate resources more effectively. These benefits are especially important when agencies face workforce shortages, fragmented systems, and increasing demands.

However, AI for Social Good is not guaranteed by good intentions. Tools built for efficiency may shift burden onto communities, reduce human contact, or make services harder to access for people with limited digital literacy. Predictive tools may label communities as risky without addressing structural causes. Chatbots may improve information access for some users while excluding people who need language, disability, or offline support.

The public health standard for social good should be rigorous. Agencies should define the public problem, identify who benefits, identify who could be harmed, involve affected communities, measure outcomes, and document safeguards. The central question is not “Can AI do this?” but “Should AI be used for this purpose, and under what conditions would it create public value?”

## Public Health Example

A health department may use AI to improve access to maternal and child health resources. A tool could help staff identify service gaps, summarize eligibility rules, translate plain-language information, and route families to local support. If designed well, the tool could reduce administrative burden and improve timely access to services.

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The same tool could fail if it assumes internet access, uses only English content, provides outdated referral information, exposes sensitive information, or replaces trusted human navigators. An AI for Social Good review would examine community needs, access barriers, language and disability accommodations, privacy, human fallback options, and measurable outcomes.

## Topic Deep Dive

AI for Social Good begins with problem definition. The problem should be a public health problem, not a technology opportunity. For example, “use a chatbot” is not a problem definition. “Reduce delays in connecting families to local nutrition, housing, transportation, and home visiting resources” is a stronger public health problem definition. Starting with the problem helps agencies avoid technology-driven solutions that do not address root needs.

Equity by design requires agencies to identify who is included, who is excluded, and who bears risk. This includes reviewing data representativeness, digital access, language access, disability access, rural access, immigration-related concerns, privacy expectations, and historical mistrust. Equity review should shape the use case, data sources, interface, outreach strategy, and evaluation metrics.

Community engagement should occur before deployment, not only after concerns arise. Affected communities can help identify priorities, unacceptable uses, preferred communication channels, needed safeguards, and measures of success. Engagement does not mean asking communities to approve a predetermined tool. It means allowing community input to change the design, scope, or decision to proceed.

Measuring social good requires more than counting users or outputs. Agencies should measure whether the AI-supported workflow improves timeliness, access, quality, fairness, staff burden, community trust, and public health outcomes. They should also monitor unintended consequences, such as reduced service quality, increased complaints, inequitable uptake, or inaccurate referrals.

Responsible innovation includes sunset and rollback planning. If an AI tool does not produce expected benefit, creates harm, or becomes outdated, the agency should be prepared to modify, pause, or retire it. Social good depends on ongoing accountability, not one-time approval.

## Risks, Failure Modes, and Guardrails

Technology-first design. A tool may be adopted because it is novel rather than because it solves a public health problem. Guardrails include problem-first planning, alternatives analysis, and public value criteria.

Unequal benefit. AI may help people with strong digital access while excluding others. Guardrails include language access, accessibility, offline options, and human fallback.

Stigma and surveillance burden. AI may label communities or increase monitoring without support. Guardrails include equity framing, community engagement, and supportive-use restrictions.

Resource diversion. AI projects may consume resources that could be used for more effective interventions. Guardrails include impact measurement, cost-benefit review, and periodic reassessment.

## Application to AI-Supported Workflows

AI for Social Good applies to every AI method. Generative AI may improve plain-language communication, but it must be accurate, accessible, and culturally appropriate. Predictive analytics may guide supportive outreach, but it should not become a tool for exclusion. RAG systems may improve staff access to guidance, but they must include current and authorized content. Agentic workflows may reduce burden, but they require boundaries and human accountability.

The strongest public health AI projects define social good in operational terms. They identify the population benefit, the equity goal, the workflow change, the safeguards, and the measurement plan. They also acknowledge tradeoffs

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and document why AI is an appropriate tool compared with non-AI alternatives.

## Reflection Questions

What public health problem would this AI use address?

Who would benefit, and who could be excluded or harmed?

How were affected communities involved in defining the need and safeguards?

What non-AI alternatives should be considered?

How will the agency measure public value, equity, trust, and unintended consequences?

## Practical Exercise

Create an AI for Social Good review brief for one public health use case. The brief should define the public health problem, intended public benefit, affected communities, equity considerations, non-AI alternatives, proposed AI role, safeguards, community engagement plan, evaluation measures, and sunset or rollback criteria.

The exercise should result in a decision-support document that helps leaders and governance reviewers determine whether the proposed AI use advances public health values.

## Expected Artifact or Evidence

AI for Social Good review brief

Public value and equity assessment

Community engagement and accessibility plan

Impact and unintended-consequence measurement plan

## Expected Assignment

- AI for Social Good review brief
- Public value and equity assessment
- Community engagement and accessibility plan
- Impact and unintended-consequence measurement plan

## Knowledge Check

- 1. What is the best definition of AI for Social Good in public health?
- 2. Why is efficiency alone not enough to show social good?
- 3. What does equity by design require?
- 4. Which use best reflects AI for Social Good?
- 5. What is strong completion evidence for this module?

## References and Resources

- WHO Ethics and Governance of Artificial Intelligence for Health: <https://www.who.int/publications/i/item/9789240029200>
- WHO Guidance on Large Multi-Modal Models: <https://www.who.int/publications/i/item/9789240084759>

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- NIST Artificial Intelligence Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
  - CDC Public Health Data Strategy: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
  - Agency equity, community engagement, accessibility, and language access policies